

Harmonic Function — Solved Questions

(Solved using the Milne-Thomson method, as in the source notes)

Method used throughout. A function $u(x, y)$ is harmonic if it satisfies Laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. If u is harmonic, its harmonic conjugate v (so that $f(z) = u + iv$ is analytic) is found by the **Milne-Thomson method**: writing $\varphi_1(x, y) = \frac{\partial u}{\partial x}$ and $\varphi_2(x, y) = \frac{\partial u}{\partial y}$, we have $f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0)$; integrating this with respect to z gives $f(z)$, and equating the imaginary part of $u + iv$ gives v .

1. Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$\begin{aligned}\therefore \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(x^3 - 3xy^2 + 3x^2 - 3y^2 + 1) \\ \Rightarrow \frac{\partial u}{\partial x} &= 3x^2 - 3y^2 + 6x = \varphi_1(x, y) \text{ (say)} (1) \\ \Rightarrow \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}(3x^2 - 3y^2 + 6x) = 6x + 6 (2)\end{aligned}$$

$$\text{Again, } \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^3 - 3xy^2 + 3x^2 - 3y^2 + 1)$$

$$\begin{aligned}\Rightarrow \frac{\partial u}{\partial y} &= -6xy - 6y = \varphi_2(x, y) \text{ (say)} (3) \\ \Rightarrow \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y}(-6xy - 6y) = -6x - 6 (4)\end{aligned}$$

Adding (2) and (4): $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + 6 - 6x - 6 = 0$. Therefore, u satisfies Laplace's equation, so u is a harmonic function.

Let v be the harmonic conjugate of u . Putting $x = z$, $y = 0$ in (1) and (3), we get $\varphi_1(z, 0) = 3z^2 + 6z$ and $\varphi_2(z, 0) = 0$. By Milne's method,

$$f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0) = 3z^2 + 6z \Rightarrow f(z) = z^3 + 3z^2 + c_1 + ic_2.$$

Then,

$$\begin{aligned}u + iv &= (x + iy)^3 + 3(x + iy)^2 + c_1 + ic_2 \\ &= x^3 + 3x^2iy - 3xy^2 - iy^3 + 3x^2 + 6xiy - 3y^2 + c_1 + ic_2 \\ &= (x^3 - 3xy^2 + 3x^2 - 3y^2) + c_1 + i(3x^2y - y^3 + 6xy + c_2)\end{aligned}$$

Equating imaginary parts, $\boxed{v = 3x^2y - y^3 + 6xy + c_2}$.

2. Show that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = 3x^2y + 2x^2 - y^3 - 2y^2$

$$\begin{aligned}\therefore \frac{\partial u}{\partial x} &= 6xy + 4x = \varphi_1(x, y) (1) & \frac{\partial^2 u}{\partial x^2} &= 6y + 4 (2) \\ \frac{\partial u}{\partial y} &= 3x^2 - 3y^2 - 4y = \varphi_2(x, y) (3) & \frac{\partial^2 u}{\partial y^2} &= -6y - 4 (4)\end{aligned}$$

Adding (2) and (4): $6y + 4 - 6y - 4 = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 4z$, $\varphi_2(z, 0) = 3z^2$. So $f'(z) = 4z - 3iz^2 \Rightarrow f(z) = 2z^2 - iz^3 + c_1 + ic_2$. Then,

$$\begin{aligned} u + iv &= 2(x + iy)^2 - i(x + iy)^3 + c_1 + ic_2 \\ &= 2x^2 + 4xy - 2y^2 - ix^3 + 3x^2y + 3ixy^2 - y^3 + c_1 + ic_2 \\ &= (3x^2y + 2x^2 - y^3 - 2y^2) + c_1 + i(4xy - x^3 + 3xy^2 + c_2) \end{aligned}$$

Equating imaginary parts, $\boxed{v = 4xy - x^3 + 3xy^2 + c_2}$.

3. Show that $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^3 + 6x^2y - 3xy^2 - 2y^3$

$$\begin{aligned} \frac{\partial u}{\partial x} &= 3x^2 + 12xy - 3y^2 = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= 6x + 12y(2) \\ \frac{\partial u}{\partial y} &= 6x^2 - 6xy - 6y^2 = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= -6x - 12y(4) \end{aligned}$$

Adding (2) and (4): $6x + 12y - 6x - 12y = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 3z^2$, $\varphi_2(z, 0) = 6z^2$. So $f'(z) = 3z^2 - 6iz^2 = 3(1 - 2i)z^2 \Rightarrow f(z) = (1 - 2i)z^3 + c_1 + ic_2$. Then,

$$\begin{aligned} u + iv &= (1 - 2i)(x + iy)^3 + c_1 + ic_2 \\ &= x^3 + 3x^2iy - 3xy^2 - iy^3 - 2ix^3 + 6x^2y + 6ixy^2 - 2y^3 + c_1 + ic_2 \\ &= (x^3 + 6x^2y - 3xy^2 - 2y^3) + c_1 + i(3x^2y - 2x^3 - y^3 + 6xy^2 + c_2) \end{aligned}$$

Equating imaginary parts, $\boxed{v = 3x^2y - 2x^3 - y^3 + 6xy^2 + c_2}$.

4. Show that $u = e^x(x \cos y - y \sin y)$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = e^x(x \cos y - y \sin y)$

$$\begin{aligned} \frac{\partial u}{\partial x} &= e^x(x \cos y - y \sin y) + e^x \cos y = \varphi_1(1) \\ \frac{\partial^2 u}{\partial x^2} &= e^x(x \cos y - y \sin y) + 2e^x \cos y(2) \\ \frac{\partial u}{\partial y} &= e^x(-x \sin y - \sin y - y \cos y) = \varphi_2(3) \\ \frac{\partial^2 u}{\partial y^2} &= e^x(-x \cos y - 2 \cos y + y \sin y)(4) \end{aligned}$$

Adding (2) and (4): $e^x(x \cos y - y \sin y + 2 \cos y - x \cos y - 2 \cos y + y \sin y) = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = ze^z + e^z$, $\varphi_2(z, 0) = 0$. So $f'(z) = ze^z + e^z \Rightarrow f(z) = ze^z - e^z + e^z + c_1 + ic_2 = ze^z + c_1 + ic_2$. Then,

$$\begin{aligned} u + iv &= (x + iy)e^{x+iy} + c_1 + ic_2 = e^x(x + iy)(\cos y + i \sin y) + c_1 + ic_2 \\ &= e^x(x \cos y - y \sin y) + c_1 + i\{e^x(x \sin y + y \cos y) + c_2\} \end{aligned}$$

Equating imaginary parts, $\boxed{v = e^x(x \sin y + y \cos y) + c_2}$.

5. Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^2 - y^2 - 2xy - 2x + 3y$

$$\begin{aligned}\frac{\partial u}{\partial x} &= 2x - 2y - 2 = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= 2(2) \\ \frac{\partial u}{\partial y} &= -2y - 2x + 3 = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= -2(4)\end{aligned}$$

Adding (2) and (4): $2 - 2 = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 2z - 2$, $\varphi_2(z, 0) = -2z + 3$. So $f'(z) = 2z - 2 - i(-2z + 3) = (2 + 2i)z - (2 + 3i)$, hence $f(z) = (1 + i)z^2 - (2 + 3i)z + c_1 + ic_2$. Then,

$$\begin{aligned}u + iv &= (1 + i)(x + iy)^2 - (2 + 3i)(x + iy) + c_1 + ic_2 \\ &= (x^2 - y^2 - 2xy) + i(x^2 - y^2 + 2xy) - 2x + 3y - i(3x + 2y) + c_1 + ic_2 \\ &= (x^2 - y^2 - 2xy - 2x + 3y) + c_1 + i(x^2 - y^2 + 2xy - 3x - 2y + c_2)\end{aligned}$$

Equating imaginary parts, $\boxed{v = x^2 - y^2 + 2xy - 3x - 2y + c_2}$.

6. Show that $u = 2x - x^3 + 3xy^2$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = 2x - x^3 + 3xy^2$

$$\begin{aligned}\frac{\partial u}{\partial x} &= 2 - 3x^2 + 3y^2 = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= -6x(2) \\ \frac{\partial u}{\partial y} &= 6xy = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= 6x(4)\end{aligned}$$

Adding (2) and (4): $-6x + 6x = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 2 - 3z^2$, $\varphi_2(z, 0) = 0$. So $f'(z) = 2 - 3z^2 \Rightarrow f(z) = 2z - z^3 + c_1 + ic_2$. Then,

$$\begin{aligned}u + iv &= 2(x + iy) - (x + iy)^3 + c_1 + ic_2 = 2x + 2iy - (x^3 + 3ix^2y - 3xy^2 - iy^3) + c_1 + ic_2 \\ &= (2x - x^3 + 3xy^2) + c_1 + i(2y - 3x^2y + y^3 + c_2)\end{aligned}$$

Equating imaginary parts, $\boxed{v = 2y - 3x^2y + y^3 + c_2}$.

7. Show that $u = e^x \cos y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = e^x \cos y$

$$\begin{aligned}\frac{\partial u}{\partial x} &= e^x \cos y = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= e^x \cos y(2) \\ \frac{\partial u}{\partial y} &= -e^x \sin y = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= -e^x \cos y(4)\end{aligned}$$

Adding (2) and (4): $e^x \cos y - e^x \cos y = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = e^z$, $\varphi_2(z, 0) = 0$. So $f'(z) = e^z \Rightarrow f(z) = e^z + c_1 + ic_2$. Then,

$$u + iv = e^{x+iy} + c_1 + ic_2 = e^x \cos y + c_1 + i(e^x \sin y + c_2).$$

Equating imaginary parts, $\boxed{v = e^x \sin y + c_2}$.

8. Can a constant function be harmonic? Why?

Solution: Yes. Let $u = c$ (a constant). Then $\frac{\partial u}{\partial x} = 0$ and $\frac{\partial u}{\partial y} = 0$, so $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 + 0 = 0$; Laplace's equation is satisfied, so a constant function is (trivially) harmonic. By the Cauchy-Riemann equations $v_y = u_x = 0$ and $v_x = -u_y = 0$, so its harmonic conjugate is also a constant.

9. Is $u = x^2 + y^2$ harmonic? Why or why not?

Solution: $\frac{\partial u}{\partial x} = 2x \Rightarrow \frac{\partial^2 u}{\partial x^2} = 2$; $\frac{\partial u}{\partial y} = 2y \Rightarrow \frac{\partial^2 u}{\partial y^2} = 2$. So $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + 2 = 4 \neq 0$. Laplace's equation is **not** satisfied, so $u = x^2 + y^2$ is **not** harmonic.

10. Is $u = x^2 - y^2$ harmonic? What is its harmonic conjugate?

Solution: $\frac{\partial u}{\partial x} = 2x = \varphi_1 \Rightarrow \frac{\partial^2 u}{\partial x^2} = 2$; $\frac{\partial u}{\partial y} = -2y = \varphi_2 \Rightarrow \frac{\partial^2 u}{\partial y^2} = -2$. Sum $= 2 - 2 = 0$, so u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 2z$, $\varphi_2(z, 0) = 0$. So $f'(z) = 2z \Rightarrow f(z) = z^2 + c_1 + ic_2$. Then $u + iv = (x + iy)^2 + c_1 + ic_2 = x^2 - y^2 + c_1 + i(2xy + c_2)$. Hence $\boxed{v = 2xy + c_2}$.

11. Show that $u = x^2 - y^2 - 4x + 6y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.

Solution: Given that, $u = x^2 - y^2 - 4x + 6y$

$$\begin{aligned} \frac{\partial u}{\partial x} &= 2x - 4 = \varphi_1(1) & \frac{\partial^2 u}{\partial x^2} &= 2(2) \\ \frac{\partial u}{\partial y} &= -2y + 6 = \varphi_2(3) & \frac{\partial^2 u}{\partial y^2} &= -2(4) \end{aligned}$$

Adding (2) and (4): $2 - 2 = 0$. So u is harmonic.

Putting $x = z$, $y = 0$: $\varphi_1(z, 0) = 2z - 4$, $\varphi_2(z, 0) = 6$. So $f'(z) = 2z - 4 - 6i \Rightarrow f(z) = z^2 - (4 + 6i)z + c_1 + ic_2$. Then,

$$\begin{aligned} u + iv &= (x + iy)^2 - (4 + 6i)(x + iy) + c_1 + ic_2 \\ &= x^2 - y^2 + 2ixy - 4x - 4iy - 6ix + 6y + c_1 + ic_2 \\ &= (x^2 - y^2 - 4x + 6y) + c_1 + i(2xy - 6x - 4y + c_2) \end{aligned}$$

Equating imaginary parts, $\boxed{v = 2xy - 6x - 4y + c_2}$.

12. Determine whether $u = 2x^2 - 4y^2 - 5xy - 4x + y$ is harmonic. If it is harmonic, find its harmonic conjugate $v(x, y)$.

Solution: Given that, $u = 2x^2 - 4y^2 - 5xy - 4x + y$

$$\begin{aligned} \frac{\partial u}{\partial x} &= 4x - 5y - 4 \Rightarrow \frac{\partial^2 u}{\partial x^2} = 4 \\ \frac{\partial u}{\partial y} &= -8y - 5x + 1 \Rightarrow \frac{\partial^2 u}{\partial y^2} = -8 \end{aligned}$$

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 4 - 8 = -4 \neq 0$. Laplace's equation is **not** satisfied, so u is **not** a harmonic function, and consequently no harmonic conjugate $v(x, y)$ exists.